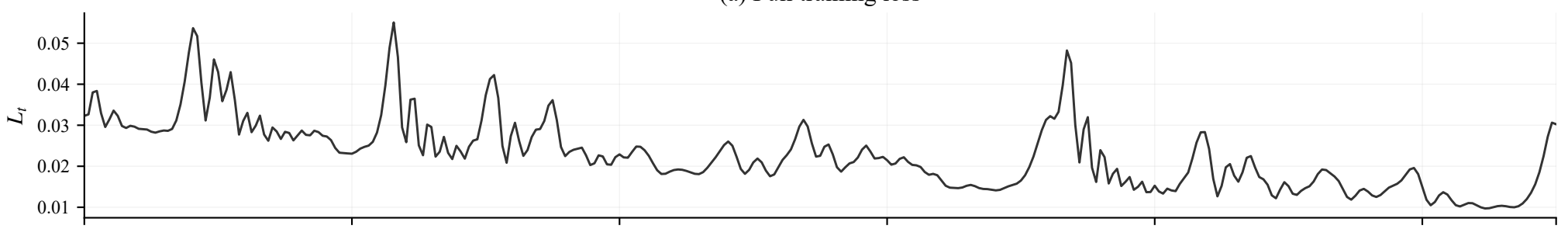
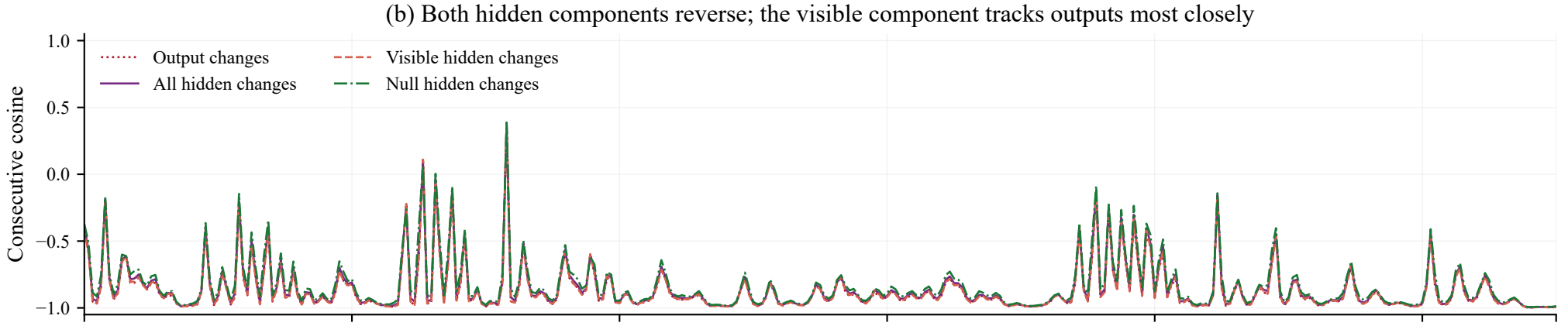
CIFAR-10 MLP: where does hidden motion concentrate?

Fixed reference head at state 448: 10 visible dimensions and 190 null dimensions per example

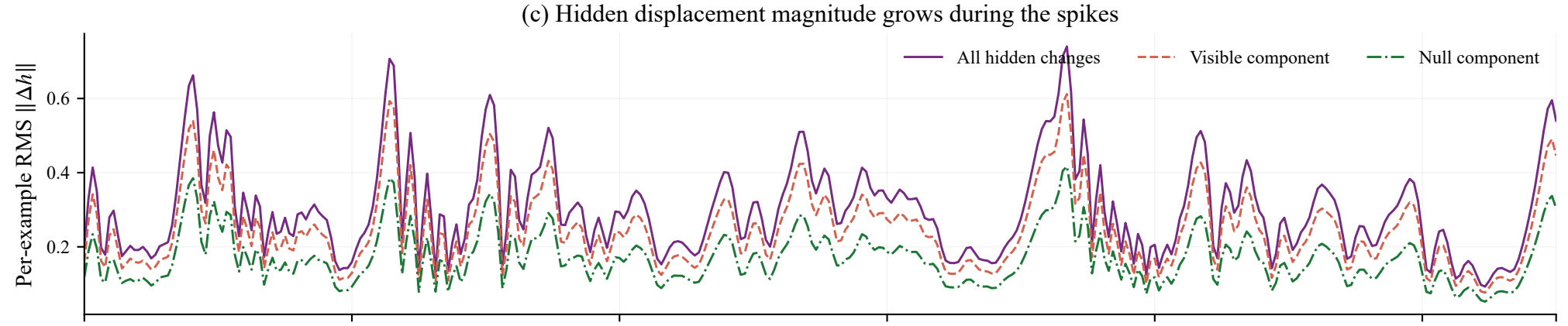
(a) Full training loss



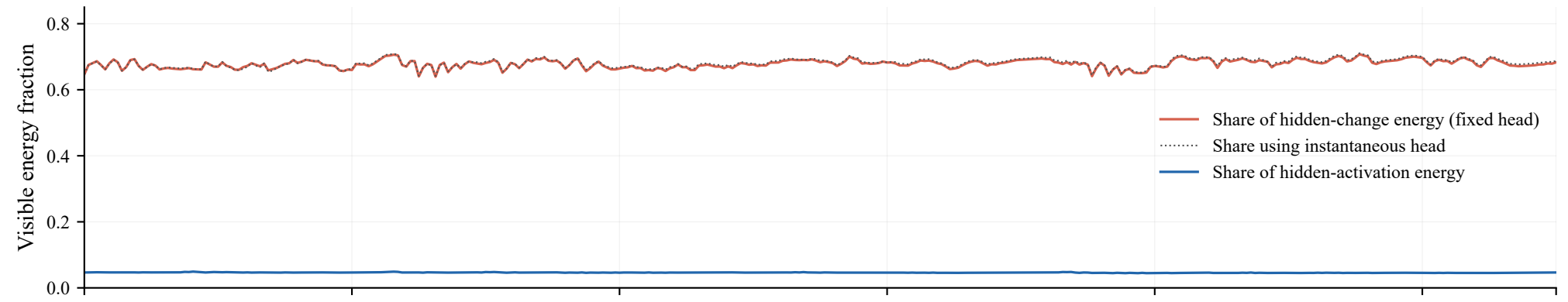
(b) Both hidden components reverse; the visible component tracks outputs most closely



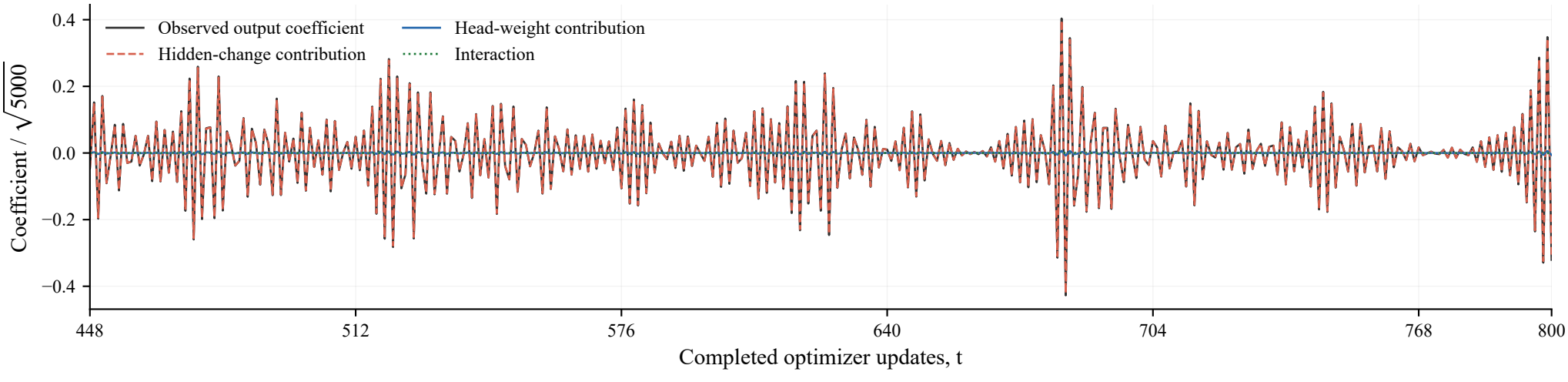
(c) Hidden displacement magnitude grows during the spikes



(d) Ten visible dimensions carry 68.1% of change energy, but only 4.5-4.9% of activation energy



(e) Leading output oscillation is supplied mainly by hidden changes through the existing head



Final hidden activation coordinate RMS stays within 0.3958-0.4009; stable magnitude does not mean stationary features.

The null component still carries 31.9% of change energy and also oscillates. Visibility is relative to the fixed reference head.

Panel (e) uses the output-space basis; bias and floating-point residuals are retained in the numerical decomposition.